MCQs for Quiz 4

1) In Bitcoin’s UTXO model, a transaction mainly does what?

A. Increases or decreases account balances directly
B. Spends old outputs and creates new outputs
C. Stores user passwords and balance history
D. Updates a miner reward table

Answer: B

Why: Bitcoin does not mainly use an account-balance model like a bank. It spends old outputs and creates new outputs.

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2) What is a UTXO?

A. A transaction that has not been confirmed
B. A miner reward that has not matured
C. An unspent transaction output
D. A script that cannot be executed

Answer: C

Why: UTXO stands for **Unspent Transaction Output**, which is a spendable piece of Bitcoin value.

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3) In a legacy Bitcoin transaction, what does scriptPubKey mainly represent?

A. The unlocking data provided by the spender
B. The locking script that defines spending conditions
C. The transaction fee field
D. The block header rule set

Answer: B

Why: scriptPubKey is the **locking script** placed in an output. It tells the network what must be satisfied to spend that output later.

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4) In a legacy Bitcoin transaction, what does scriptSig mainly provide?

A. The previous block hash and nonce
B. A Merkle proof
C. The unlocking data, usually signature and public key
D. The new output values

Answer: C

Why: This matches the standard legacy transaction idea and also matches the style of your earlier quiz pool.

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5) In simple terms, Bitcoin checks which of the following before allowing spending?

A. If the miner likes the transaction
B. If scriptPubKey accepts the unlocking data
C. If the transaction value is above 1 BTC
D. If the sender has a bank account

Answer: B

Why: The lecture phrase is basically: if the locking condition is satisfied by the unlocking data, then the spend is valid and new outputs can be created.

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6) Why might a Bitcoin transaction use multiple inputs?

A. Because one UTXO may not be enough to cover the payment
B. Because every receiver must create one input
C. Because SegWit requires at least two inputs
D. Because transaction fees can only be paid through extra inputs

Answer: A

Why: Multiple inputs let a user combine earlier unspent outputs to make a larger payment.

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7) Why might a Bitcoin transaction create multiple outputs?

A. Because every transaction must pay at least three users
B. Because one output is for mining and one for validation
C. Because it may pay a recipient and also return change to the sender
D. Because outputs are required to match the number of inputs

Answer: C

Why: Very common pattern: payment output plus change output.

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8) What is the main use of OP_RETURN in Bitcoin?

A. To make an output spendable by anyone
B. To intentionally create an unspendable output, often for data
C. To reduce mining difficulty
D. To sign a SegWit witness

Answer: B

Why: OP_RETURN outputs are provably unspendable and are typically used to attach small pieces of data.

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9) Why is the value of an OP_RETURN output usually zero or negligible?

A. Because Bitcoin does not allow value in outputs
B. Because miners ignore such outputs
C. Because locking valuable BTC in an unspendable output would waste it
D. Because OP_RETURN can only be used in block headers

Answer: C

Why: If the output can never be spent, putting valuable BTC there would trap it forever.

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10) Legacy Bitcoin signatures are created using:

A. RSA
B. AES
C. ECDSA
D. SHA-256 only

Answer: C

Why: The lecture explicitly says legacy signatures use ECDSA.

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11) An ECDSA signature mainly consists of:

A. nonce and block hash
B. r and s
C. value and index
D. public key and txid

Answer: B

Why: The lecture directly states that an ECDSA signature is a pair ((r,s)).

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12) In the ECDSA signature format shown in the lecture, the final extra byte indicates:

A. the block version
B. the public key length
C. which parts of the transaction were hashed and signed
D. whether the transaction is SegWit

Answer: C

Why: That is the role of the **sighash byte**.

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13) What does the sighash byte do?

A. It decides the block reward
B. It defines the scope of the transaction data covered by the signature
C. It prevents double SHA-256
D. It identifies the miner who first saw the transaction

Answer: B

Why: Same idea: it specifies what exactly is signed.

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14) Why do leading zero bytes in legacy signature encoding matter?

A. They can make the transaction invalid immediately
B. They can change byte representation while keeping signature meaning valid
C. They store the output index
D. They show the number of confirmations

Answer: B

Why: This is the setup for transaction malleability. Same logical authorization, but different byte representation.

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15) Transaction malleability means:

A. a transaction can spend different inputs after confirmation
B. the transaction fee can increase automatically
C. the txid can change without changing the actual spending effect
D. the block header can be edited after mining

Answer: C

Why: This is one of the core Week 6 definitions.

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16) Why was txid malleability possible in legacy Bitcoin?

A. Because miners manually rename transactions
B. Because txid calculation included scriptSig and signature bytes

C. Because outputs were not hashed
D. Because block headers ignored transaction order

Answer: B

Why: In legacy Bitcoin, scriptSig was part of the serialized transaction used to compute txid.

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17) Which of the following can happen because of transaction malleability?

A. Wallet confusion about which txid got confirmed
B. Total BTC supply changing
C. Previous block hash becoming invalid
D. Merkle roots being optional

Answer: A

Why: Wallets may expect one txid but see a different equivalent version confirmed.

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18) Why is transaction malleability especially harmful for Layer-2 systems?

A. They do not use digital signatures
B. They depend on stable, known txids for dependent transactions
C. They remove all outputs
D. They use account balances instead of UTXOs

Answer: B

Why: This is a very important conceptual question. Payment channels and similar systems often need to know the exact earlier txid in advance.

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19) SegWit stands for:

A. Segmented Window Transactions
B. Serialized Global Witness

C. Segregated Witness
D. Signature Grouped With Inputs

Answer: C

Why: Standard definition from the lecture.

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20) What problem was SegWit mainly introduced to solve in these lectures?

A. Excessive miner rewards
B. Transaction malleability
C. Lack of timestamps
D. Too many public keys in blocks

Answer: B

Why: That is the main focus of the Week 6 SegWit discussion.

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21) What is the main idea of SegWit?

A. Put outputs into the block header
B. Separate witness data from the part used to compute txid
C. Remove signatures entirely
D. Replace SHA-256 with RSA

Answer: B

Why: This is probably the most important one-line summary of SegWit.

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22) In SegWit, the witness mainly contains:

A. mining reward data
B. previous block hashes
C. proof data like signatures and public keys
D. transaction fees only

Answer: C

Why: Witness is the authorization proof moved outside the legacy scriptSig position.

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23) In a standard P2WPKH SegWit transaction, scriptSig is usually:

A. full of opcodes and signature
B. used to store the Merkle root
C. empty
D. equal to the block hash

Answer: C

Why: In SegWit P2WPKH, unlocking data is in the witness, so scriptSig is empty.

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24) P2WPKH stands for:

A. Pay to Whole Public Key Header
B. Pay to Witness Public Key Hash
C. Proof to Witness Public Key Hash
D. Pay to Wrapped Public Key Hash

Answer: B

Why: Exact lecture term.

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25) Compared to legacy P2PKH, P2WPKH mainly changes:

A. who is allowed to spend
B. where the unlocking data is stored
C. the total supply of bitcoin
D. the value field format

Answer: B

Why: Same ownership idea, different structure. The proof data moves to witness.

26) In visible SegWit P2WPKH output form, why do there appear to be “no opcodes”?

A. Because SegWit removes all validation logic
B. Because validation is done by miners manually
C. Because Bitcoin internally expands it into a virtual P2PKH-like script
D. Because old nodes reject it

Answer: C

Why: This is a classic tricky conceptual point. The visible output is compact, but Bitcoin reconstructs equivalent checking logic internally.

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27) In SegWit P2WPKH validation, Bitcoin still ultimately checks:

A. whether the miner owns the output
B. whether the provided public key hashes correctly and signature verifies
C. whether the block has more than one transaction
D. whether OP_RETURN exists in the same block

Answer: B

Why: SegWit changed storage location, not the fundamental ownership test.

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28) In the P2WPKH execution sequence, the final successful stack result is:

A. 0
B. OP_RETURN
C. TRUE
D. 21 million

Answer: C

Why: The lecture sequence ends with True when validation succeeds.

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29) Multisig means:

A. one signature verified twice
B. one miner signs for all users
C. multiple signatures are required to authorize spending
D. multiple blocks share the same header

Answer: C

Why: That is the basic meaning of multisig.

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30) In a 2-of-3 multisig setup, spending requires:

A. exactly 3 signatures
B. any 2 valid signatures out of 3 allowed keys
C. only the oldest key
D. one signature and one nonce

Answer: B

Why: M-of-N means M required signatures out of N possible keys.

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31) Which is a real use case of multisig from the lecture?

A. Changing mining difficulty
B. Shared accounts and escrow
C. Compressing block headers
D. Preventing Merkle roots

Answer: B

Why: The lecture gives shared accounts, escrow, security, and Lightning channels as examples.

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32) In multisig notation, N represents:

A. the number of required signatures
B. the total number of authorized keys
C. the number of outputs
D. the number of blocks waited

Answer: B

Why: M-of-N means M required out of N total keys.

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33) In multisig notation, M represents:

A. number of miners validating the block
B. number of required signatures
C. number of Merkle proofs
D. number of change outputs

Answer: B

Why: Straight definition.

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34) P2WSH stands for:

A. Pay to Witness Script Hash
B. Pay to Whole Script Header
C. Proof to Witness Script Hash
D. Pay to Wrapped SegWit Hash

Answer: A

Why: Exact lecture wording.

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35) Why is P2WSH useful?

A. It locks outputs to the hash of a whole script, allowing complex spending rules
B. It removes the need for public keys
C. It stores txids in the block header
D. It prevents mining rewards from halving

Answer: A

Why: P2WSH is used for more complex conditions like multisig.

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36) In P2WSH spending, the last witness item must contain:

A. the previous block hash
B. the witness script
C. the miner signature
D. the Merkle root

Answer: B

Why: The lecture explicitly mentions that the last witness item contains the witness script.

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37) Before executing a P2WSH multisig script, Bitcoin first checks:

A. whether the current block time is exactly synchronized
B. whether SHA256 of the revealed witness script matches the output commitment
C. whether the transaction has exactly two outputs
D. whether the signature is RSA-based

Answer: B

Why: This is one of the most important execution steps. The revealed script must hash to the committed hash in the old output.

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38) Why is a dummy OP_0 included in multisig witness data?

A. To reduce fees
B. To signal Taproot
C. Because of the historical CHECKMULTISIG bug
D. To represent the number of outputs

Answer: C

Why: Very likely short conceptual trap. The extra zero is only there because of the bug.

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39) In multisig validation, success occurs when:

A. the number of valid signatures equals or exceeds the required M
B. any one key is present
C. the transaction has no outputs
D. the public keys are alphabetical

Answer: A

Why: Enough valid signatures must match the allowed key set.

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40) A Bitcoin block header has how many fields according to the lecture?

A. 4
B. 5
C. 6
D. 8

Answer: C

Why: The lecture clearly lists six fields.

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41) Which of the following is not a block header field?

A. Previous block hash
B. Merkle root
C. Nonce
D. scriptSig

Answer: D

Why: scriptSig belongs to transactions, not the block header.

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42) Which block header field links the current block to its parent?

A. Version
B. Previous block hash
C. Timestamp
D. Nonce

Answer: B

Why: That field is what makes blocks form a chain.

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43) Which block header field summarizes the transactions inside the block?

A. Previous block hash
B. Merkle root hash
C. Version
D. Nonce

Answer: B

Why: Merkle root is the compact fingerprint of all block transactions.

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44) Which block header field is repeatedly changed during mining?

A. Merkle root
B. Nonce
C. Version
D. Output index

Answer: B

Why: Miners keep changing the nonce while searching for a valid hash.

45) The block header hash in Bitcoin is computed using:

A. SHA256 once
B. SHA256 twice
C. RSA then SHA256
D. ECDSA twice

Answer: B

Why: Double SHA-256.

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46) Why does the lecture question whether Bitcoin base layer is ideal for buying coffee?

A. Because coffee vendors do not use cryptography
B. Because Bitcoin cannot represent small amounts
C. Because on-chain confirmation is slow, public, and limited in throughput
D. Because blocks do not store transactions

Answer: C

Why: This is the big motivation for micropayment ideas.

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47) Which of the following is a practical concern for coffee payments on Bitcoin base layer?

A. Need to wait around 10 minutes on average for a block
B. Bitcoin uses no timestamps
C. Outputs cannot hold small values
D. Public keys are secret

Answer: A

Why: Block time is a major practical issue.

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48) Which of the following is another coffee-payment concern mentioned in the lecture?

A. Every coffee purchase being publicly broadcast
B. Coffee cannot be priced in sats
C. Miners choose beverage types
D. Only multisig outputs can buy coffee

Answer: A

Why: Privacy is one of the concerns raised.

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49) Bitcoin Script can support which of the following according to Lecture 13?

A. Signature checks, branching, multisig, time locks
B. Only signature checks and nothing else
C. Only database queries
D. Automatic fiat conversion

Answer: A

Why: The lecture explicitly lists all of these categories.

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50) `OP_CHECKMULTISIG` is used to:

A. summarize transactions into a block header
B. require multiple approvals/signatures
C. calculate the nonce
D. convert BTC to fiat

Answer: B

Why: Standard script capability from the lecture.

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51) `OP_CHECKLOCKTIMEVERIFY` is related to:

A. relative or absolute spending restriction by time/block condition
B. Merkle tree calculation
C. public key compression
D. mining subsidy schedule

Answer: A

Why: CLTV is for time locking funds.

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52) Miniscript is compared in the lecture to:

A. SQL for Bitcoin
B. TypeScript for Bitcoin
C. Java for Ethereum
D. HTML for miners

Answer: B

Why: Exact analogy in the lecture.

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53) Why is Miniscript useful?

A. It removes the need for Bitcoin Script
B. It makes Bitcoin mining faster
C. It makes complex spending policies easier to read and reason about
D. It changes BTC issuance rules

Answer: C

Why: Raw Script is low-level and hard to verify, while Miniscript is more structured.

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54) In the UTXO model, each output must mention:

A. only a transaction ID
B. value and scriptPubKey
C. witness and nonce
D. block height and Merkle proof

Answer: B

Why: This is directly stated in the lecture.

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55) Each input in a Bitcoin transaction must reference:

A. exactly one previous output
B. all outputs of the previous block
C. exactly two previous outputs
D. one miner reward and one fee output

Answer: A

Why: Core UTXO rule.

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56) In SegWit transactions, scriptSig is empty because:

A. signatures are no longer needed
B. the witness field contains the unlocking data instead
C. the miner signs on behalf of the sender
D. outputs are not locked anymore

Answer: B

Why: Important contrast with legacy.

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57) For a valid Bitcoin transaction, the sum of input values must be:

A. exactly equal to sum of outputs always
B. greater than or equal to sum of outputs
C. less than outputs if SegWit is used
D. unrelated to outputs

Answer: B

Why: If inputs are larger than outputs, the difference becomes fee.

58) Why does Bitcoin not require user accounts in the UTXO model?

A. Because it tracks spendable outputs instead of account rows
B. Because public keys act as usernames
C. Because miners keep balances privately
D. Because only exchanges use accounts

Answer: A

Why: The lecture explicitly emphasizes this.

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59) The parent transaction is also called the:

A. witness transaction
B. funding transaction
C. rollback transaction
D. script transaction

Answer: B

Why: The lecture uses parent = funding transaction terminology.

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60) The child transaction is also called the:

A. signature transaction
B. spending transaction
C. header transaction
D. coinbase transaction

Answer: B

Why: Parent creates the value, child spends it.

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61) In transaction chaining, Bitcoin transactions form:

A. only isolated independent records
B. a chain or graph of spending relationships
C. random unordered fee tables
D. a sequence of account resets

Answer: B

Why: The lecture emphasizes that transactions also form a chain of spending, not just blocks.

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62) Why do nodes keep a UTXO database such as chainstate?

A. To store all forgotten private keys
B. To speed up validation of new transactions
C. To reduce block size to zero
D. To replace the blockchain entirely

Answer: B

Why: Otherwise every validation would need expensive full-history searching.

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63) The UTXO database stores:

A. all historical outputs ever created
B. only currently unspent outputs
C. only SegWit outputs
D. only miner rewards

Answer: B

Why: Only still-spendable outputs are needed there.

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64) Which of the following would make validation fail under the simplified algorithm?

A. Referenced parent UTXO not found in the DB
B. Inputs larger than outputs

C. Presence of a change output
D. Use of P2WPKH

Answer: A

Why: If the referenced UTXO is not there, it cannot be spent.

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65) Another validation failure case is when:

A. the provided signature does not match the spending condition of the parent UTXO
B. the transaction has two outputs
C. the transaction uses a public key
D. the parent transaction had multiple outputs

Answer: A

Why: Authorization must match the owner/spending rule.

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66) If a transaction is valid, nodes update the UTXO database by:

A. keeping old parent UTXOs and ignoring child outputs
B. removing spent parent UTXOs and adding new child outputs
C. deleting the entire chainstate
D. changing account balances only

Answer: B

Why: This is exactly how Bitcoin state moves forward.

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67) A change output is needed because:

A. miners demand a special separate fee output
B. Bitcoin consumes UTXOs as whole pieces, so leftover value must be explicitly returned
C. public keys expire after one use
D. SegWit requires two outputs minimum

Answer: B

Why: Very common trap. Bitcoin does not “partially spend” an output in place. It spends the whole old output and creates new ones.

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68) If Alice spends a 1.5 BTC input but creates only a 0.5 BTC output to Bob and no change output, then the remaining value becomes:

A. invalid and disappears from the system
B. automatically returned to Alice
C. miner fee
D. block reward

Answer: C

Why: Valid, but economically terrible.

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69) A collaborative transaction is one where:

A. multiple parties agree on inputs/outputs and add their signatures
B. miners collaboratively rewrite past blocks
C. multiple blocks share the same txid
D. two accounts merge automatically

Answer: A

Why: This idea becomes important for payment channels and shared control.

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70) PSBT stands for:

A. Public Script Broadcast Transaction
B. Partially Signed Bitcoin Transaction
C. Private SegWit Block Transfer
D. Parent Script Balance Table

Answer: B

Why: Exact term from the lecture notes.

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71) A PSBT is:

A. already fully final and broadcastable
B. a transaction with some but not all required signatures yet
C. a miner’s reward transaction
D. a block header with missing nonce

Answer: B

Why: It is an in-progress transaction package.

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72) 1 BTC equals:

A. 10^6 sats
B. 10^7 sats
C. 10^8 sats
D. 10^9 sats

Answer: C

Why: Standard unit fact used in the micropayment discussion.

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73) Why does the lecture say to “start thinking in satoshis”?

A. Because Bitcoin blocks are counted in sats
B. Because small payments feel more practical when expressed in sats instead of tiny BTC decimals
C. Because miners accept only sats in headers
D. Because SegWit requires sat-based scripts

Answer: B

Why: It is a mental-unit framing issue.

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74) A key reason on-chain Bitcoin struggles with micropayments is that transaction fees depend mainly on:

A. transaction value in dollars
B. number of confirmations desired
C. transaction size in bytes
D. public key type only

Answer: C

Why: This is one of the biggest Week 7 points.

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75) Because fee depends on size, a $1 Bitcoin payment can pay roughly the same fee as a $100,000 payment if:

A. they are broadcast by the same wallet
B. they have similar transaction byte size
C. they are mined in the same block
D. they use the same public key

Answer: B

Why: Very quiz-worthy and slightly tricky. Miners care about block space, not the dollar amount being transferred.

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76) The main idea for reducing micropayment fees is to:

A. remove signatures from Bitcoin
B. use one on-chain funding transaction and many off-chain balance updates
C. store all coffee purchases in OP_RETURN
D. wait for fees to become zero forever

Answer: B

Why: This is the central payment-channel idea.

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77) A funding transaction in the payment-channel discussion is:

A. the final settlement transaction only
B. the initial on-chain transaction that locks funds into the channel
C. a miner’s subsidy output
D. an invalid old commitment

Answer: B

Why: It opens the shared channel.

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78) A commitment transaction mainly represents:

A. the current balance split inside the channel
B. the current network difficulty
C. a Merkle proof of inclusion
D. the previous block hash

Answer: A

Why: Commitment transactions track current agreed state off-chain.

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79) In the lecture’s micropayment idea, why are commitment transactions useful?

A. They allow many payment updates without putting each one on-chain
B. They replace the block header
C. They eliminate all signatures
D. They mint new BTC for coffee vendors

Answer: A

Why: They reduce repeated on-chain fees.

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80) Which of the following best describes a payment channel?

A. A setup that lets two parties update balances off-chain and settle later on-chain
B. A miner-only communication line
C. A block header compression method
D. A new consensus algorithm

Answer: A

Why: That is the lecture's core definition.

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81) Why was SegWit especially important for payment channels?

A. It made txids more stable by removing witness data from the txid-critical part
B. It increased total Bitcoin supply
C. It removed the need for multisig
D. It made blocks larger than needed

Answer: A

Why: This connects Week 6 and Week 7 beautifully. Channels need reliable txids.

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82) Legacy transaction format is problematic for payment channels because:

A. it does not support outputs
B. signatures inside txid-related serialization can be malleated
C. it uses Merkle roots
D. it stores timestamps

Answer: B

Why: That is the exact reason legacy txids were dangerous for dependent transactions.

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83) In the lecture, which output type is commonly associated with a simple SegWit user payment?

A. P2WSH
B. P2WPKH
C. OP_RETURN
D. coinbase only

Answer: B

Why: P2WPKH is the SegWit version of “pay to public key hash” style ownership.

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84) Which output type is commonly used for more complex policies like multisig?

A. P2WSH
B. P2WPKH
C. OP_RETURN
D. P2PKH only

Answer: A

Why: Because it locks to a whole script hash.

---

85) Which statement is correct about SegWit and old nodes, according to the broader notes?

A. Old nodes fully execute witness programs exactly like new nodes
B. Old nodes reject all SegWit transactions immediately
C. New nodes understand witness rules, while old nodes do not interpret witness the same way
D. Old nodes compute witness Merkle trees for block headers

Answer: C

Why: This is a slightly trickier conceptual one from the extended notes. New nodes know how to process the witness program properly.

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86) Which of the following best describes the purpose of the Merkle root in the block header?

A. It stores all scripts directly
B. It summarizes the transaction set inside the block
C. It determines transaction fees
D. It replaces the nonce

Answer: B

Why: Merkle root is the compact summary of all transactions.

---

87) Which of the following is the best short description of witness?

A. A miner reward calculator
B. Proof data showing the spender is authorized
C. A list of all outputs in the block
D. A database of lost coins

Answer: B

Why: Nice short-definition MCQ.

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88) If two valid byte encodings of the same legacy signature exist, then:

A. they must always produce the same txid
B. they can produce different txids while keeping the same economic effect
C. both become invalid automatically
D. the Merkle root becomes optional

Answer: B

Why: This is basically the essence of transaction malleability.

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89) Which of the following is most closely tied to the CHECKMULTISIG bug?

A. extra dummy OP_0 item
B. empty scriptSig in P2WPKH
C. Merkle tree generation
D. block timestamp tolerance

Answer: A

Why: Direct association.

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90) Which statement best compares P2PKH and P2WPKH?

A. P2PKH uses witness, P2WPKH uses scriptSig
B. P2PKH and P2WPKH have the same basic ownership idea, but P2WPKH stores unlocking proof in witness
C. P2WPKH removes the need for signatures
D. P2PKH is multisig while P2WPKH is single-sig only

Answer: B

Why: Great comparison question.

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91) Which statement about transaction fees is correct?

A. Fees are always proportional to BTC value sent
B. Fees are based mainly on transaction byte size
C. Fees are stored as a separate compulsory output
D. Fees exist only in SegWit

Answer: B

Why: Very important Week 7 point.

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92) Which of the following best explains why Bitcoin base layer alone is not ideal for billions of coffee purchases per day?

A. Bitcoin cannot represent small units
B. Bitcoin cannot store outputs
C. Throughput is limited and each on-chain transaction consumes scarce block space
D. ECDSA only works for large payments

Answer: C

Why: The lecture directly raises the scalability concern.

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93) The lecture's "coffee problem" is mainly used to motivate:

A. miner centralization
B. micropayment platforms and payment channels
C. removing public keys
D. abandoning UTXOs

Answer: B

Why: That is the transition into off-chain balance updates.

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94) In the UTXO model, the exact coin being spent is identified by:

A. txid only
B. output index only
C. previous txid plus output index
D. public key hash only

Answer: C

Why: This is fundamental. txid alone is not enough; you need the specific output index too. This idea appears strongly in the UTXO discussions around spending specific outputs.

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95) Which of the following best describes the role of a child transaction?

A. It creates the original locked coin
B. It spends an output created by a parent transaction
C. It stores the Merkle root
D. It replaces the block header

Answer: B

Why: Parent funds, child spends.

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96) If the total value of referenced parent UTXOs is less than the total value of a child transaction's outputs, then the transaction is:

A. valid with zero fee
B. invalid
C. automatically fixed by miners
D. valid only in SegWit

Answer: B

Why: You cannot create more value than you consume.

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97) Which of the following is the best one-line explanation of SegWit from these lectures?

A. It moved miners into the witness field
B. It fixed txid malleability by separating witness data from the txid-critical transaction body
C. It removed block headers from Bitcoin
D. It made all scripts multisig

Answer: B

Why: Very strong exam-style one-liner.

98) Which of the following is the best one-line explanation of multisig?

A. One transaction with two fees
B. A spending policy requiring multiple signatures rather than one
C. A block with multiple Merkle roots
D. A new mining algorithm

Answer: B

Why: Straight concept definition.

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99) Which of the following is the best one-line explanation of a payment channel?

A. A way to put all payments permanently into OP_RETURN
B. A two-party setup for making many payments off-chain and settling later on-chain
C. A miner-only fee negotiation mechanism
D. A replacement for the UTXO model

Answer: B

Why: Strong final concept check.

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100) Which pairing is correct?

A. scriptPubKey = unlocking data, scriptSig = locking condition
B. scriptPubKey = locking condition, scriptSig = unlocking data in legacy
C. scriptPubKey = Merkle proof, scriptSig = block header hash
D. scriptPubKey = block reward, scriptSig = nonce

Answer: B

Why: This is one of the most likely basic-concept MCQs.

Tricky mixed MCQs

101) Which statement is false?

A. SegWit P2WPKH keeps scriptSig empty and moves unlocking data to witness
B. Multisig can be expressed as M-of-N
C. Transaction fee mainly depends on byte size, not payment value
D. In Bitcoin, every payment must directly update an account balance

Answer: D

Why: This is false because Bitcoin uses the UTXO model, not direct account balance updates.

---

102) Which statement is false?

A. OP_RETURN outputs are usually intended for future spending
B. Transaction malleability was a serious issue for dependent off-chain transactions
C. P2WSH locks to the hash of a script
D. Block header contains Merkle root

Answer: A

Why: OP_RETURN outputs are intentionally unspendable.

---

103) Which statement is correct?

A. SegWit removes signatures from Bitcoin completely
B. SegWit changes where proof data is stored, not the need for ownership verification
C. SegWit makes scriptPubKey disappear from outputs
D. SegWit stores everything inside the block header

Answer: B

Why: This is a subtle conceptual trap. SegWit does not remove authorization logic. It changes structure.

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104) Which statement is correct?

A. In legacy Bitcoin, changing valid signature encoding could not affect txid
B. In legacy Bitcoin, txid was independent of scriptSig
C. In legacy Bitcoin, changing valid signature byte representation could change txid
D. Legacy Bitcoin used witness to protect txid

Answer: C

Why: Classic malleability point.

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105) Which statement is correct?

A. A change output is optional only when no leftover value needs to return
B. A change output is always mandatory even when sending the full amount minus fee
C. Change outputs exist only in multisig
D. Change outputs are stored in the block header

Answer: A

Why: If the payer transfers the entire input value minus fee to the payee, then no change output is needed. This also matches the style seen in your older quiz pool.

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106) Which statement is incorrect?

A. Witness is proof data for spending
B. chainstate stores currently unspent outputs
C. Nonce is a field in the block header
D. Merkle root is stored in scriptSig

Answer: D

Why: Merkle root is stored in the block header, not in scriptSig.

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107) Which statement is most accurate?

A. Payment channels reduce fees by making every small payment a separate on-chain transaction

B. Payment channels reduce fees by shifting many balance updates off-chain between opening and closing
C. Payment channels work only if txids are unstable
D. Payment channels eliminate the need for signatures

Answer: B

Why: Good summary of the whole micropayment idea.

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108) Which statement is most accurate?

A. P2WPKH and P2WSH are identical
B. P2WPKH is generally for pubkey-hash style spending, while P2WSH is for full-script-hash style spending
C. P2WSH has no witness
D. P2WPKH is used only for block headers

Answer: B

Why: This comparison is very important.

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109) Which statement is most accurate?

A. A transaction with inputs greater than outputs is always invalid
B. A transaction with inputs greater than outputs is valid, and the difference becomes fee
C. A transaction with inputs greater than outputs must automatically create change
D. Such transactions are valid only in legacy but not SegWit

Answer: B

Why: Subtle but essential. Change output is common, but not automatic. If you omit it, leftover becomes fee.

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110) Which statement is most accurate?

A. Bitcoin base layer is perfect for daily micropayments because fee scales with payment value
B. Bitcoin base layer struggles with micropayments because fee depends on size and confirmations are slow

C. Bitcoin cannot represent tiny value at all
D. Bitcoin has no off-chain solutions

Answer: B

Why: Very strong integrated concept question.

Ultra-short rapid fire set

111) Legacy unlocking data lives in:

A. witness
B. scriptSig
C. block header
D. Merkle root

Answer: B

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112) SegWit unlocking data lives in:

A. witness
B. nonce
C. timestamp
D. version

Answer: A

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113) Malleability mainly affected:

A. stable txids
B. block size only
C. satoshi denomination
D. Merkle tree existence

Answer: A

114) P2WSH commits to:

A. public key hash only
B. script hash
C. block hash
D. coinbase reward

Answer: B

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115) Multisig 2-of-2 means:

A. either one can sign
B. both must sign
C. 2 out of any 3 must sign
D. no signature needed

Answer: B

---

116) Block header mining target is tied to:

A. difficulty
B. scriptSig
C. change output
D. sighash type

Answer: A

---

117) chainstate is mainly for:

A. storing private keys
B. faster validation
C. mining rewards
D. wallet UI

Answer: B

118) Change output usually goes to:

A. miner
B. sender
C. anyone-can-spend
D. block header

Answer: B

119) Funding transaction is usually:

A. the opening on-chain transaction for a channel
B. the final block reward
C. the Merkle proof transaction
D. the OP_RETURN transaction

Answer: A

120) Commitment transactions are usually:

A. off-chain state updates
B. block headers
C. witness-only blocks
D. coinbase records

Answer: A